| What | Linear Algebra | Fourier Series |
|---|---|---|
| **Vector Space** | $\mathbb{R}^n$ | Piecewise smooth $2\pi$-periodic functions on $\mathbb{R}$ |
| **Inner Product** | $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} u_i v_i$ | $\langle f(t), g(t) \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) g(t) dt$ |
| **Orthonormal Basis** $\left(\mathbb{R}^3\right)$ | $\{(1,0,0),(0,1,0),(0,0,1)\}$ | $\{1, \cos nt, \sin nt\}$ |
| **Representation of a Vector in the Basis** | $\mathbf{x} = \sum_{i=1}^{n} a_i \mathbf{e}_i$ | $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$ |
| **Coefficients are Projections** | $a_i = \langle \mathbf{x}, \mathbf{e}_i \rangle$ | $a_0 = \langle f(t), 1 \rangle$<br>$a_m = \langle f(t), \cos mt \rangle$<br>$b_m = \langle f(t), \sin mt \rangle$ |

Table 1: Vector Spaces: Linear Algebra vs. Fourier Series

Note: Some people think that the orthonormal basis for the Fourier series should be $\{1, \sqrt{2}\cos nt, \sqrt{2}\sin nt\}$, since $\langle 1, 1 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} 1^2 \, dt = \frac{1}{\pi} t \Big|_{-\pi}^{\pi} = \frac{1}{\pi} \left(\pi - (-\pi)\right) = \frac{1}{\pi} 2\pi = 2$. OTOH, $\langle \cos nt, \cos nt \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos^2 nt \, dt = \frac{1}{\pi} \pi = 1$.

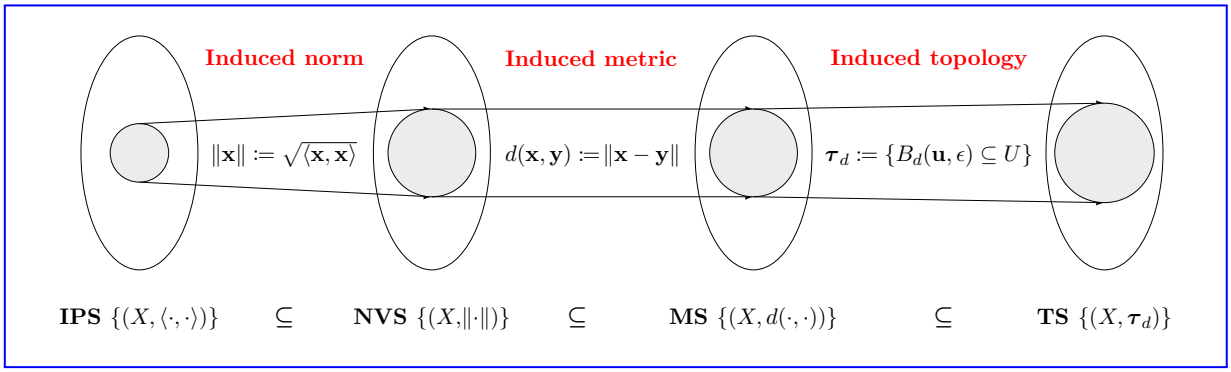


Figure 1: Induced Norm, Metric, and Topology

| | | | |
|---|---|---|---|
| **IPS** | $--$ | Inner Product Spaces | $(X, \langle \mathbf{x}, \mathbf{y} \rangle)$ |
| **NVS** | $--$ | Normed Vector Spaces | $(X, \|\mathbf{x}\|) = (X, \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle})$ |
| **MS** | $--$ | Metric Spaces | $(X, d(\mathbf{x}, \mathbf{y})) = (X, \|\mathbf{x} - \mathbf{y}\|) = (X, \sqrt{\langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle})$ |
| **TS** | $--$ | Topological Spaces | $(X, \boldsymbol{\tau}_d)$, where $\boldsymbol{\tau}_d = \{\text{open subsets of } (X, d)\}$ |

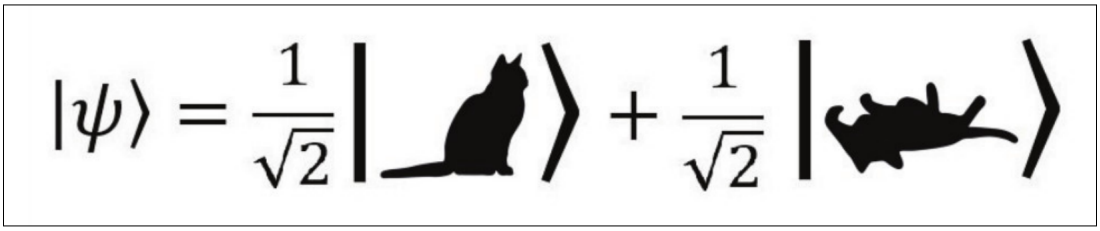


Figure 2: Quantum superposition and Schrödinger's cat

$n=\langle\ ,\ \rangle$
$G(\ )$
$\ \ \ \

0.6249999121401877

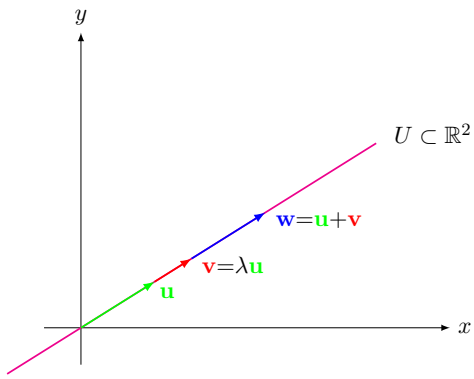


Figure 5: The line $U$ is a linear subspace of $\mathbb{R}^2$

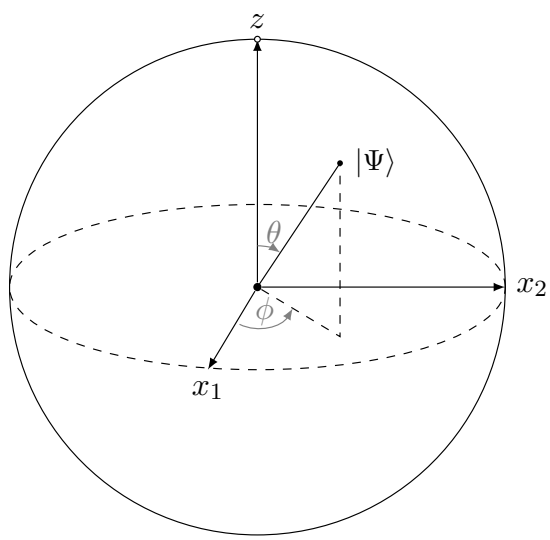


Figure 6: The Bloch Sphere

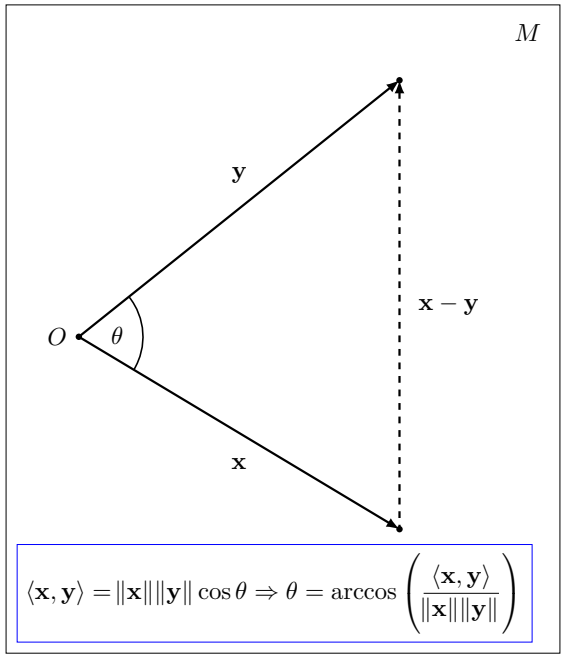


Figure 7: Geometric interpretation of $\langle \mathbf{x}, \mathbf{y} \rangle$

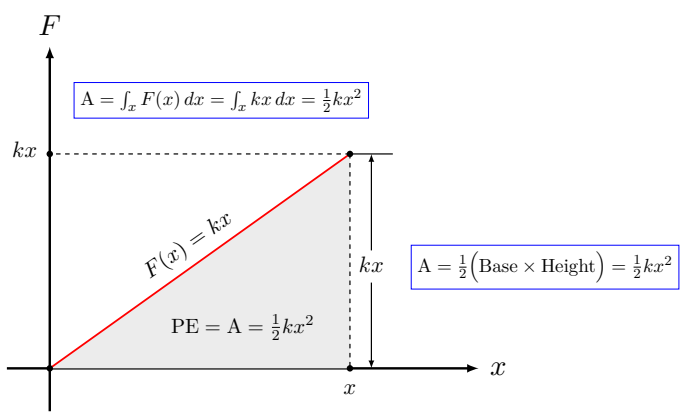


Figure 8: Potential Energy of a Simple Harmonic Oscillator

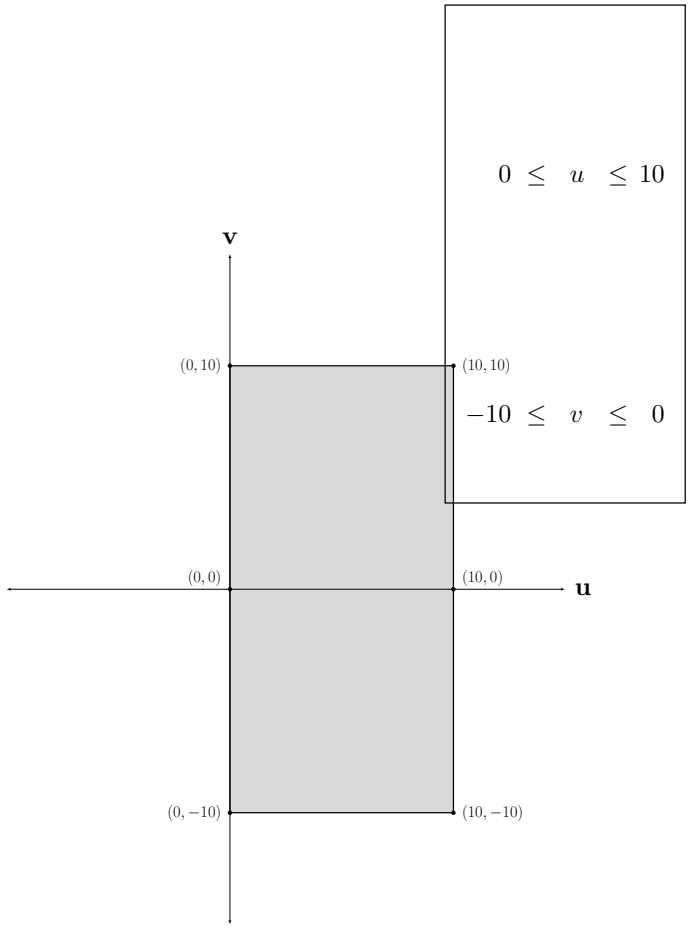


Figure 9: The transformed parallelogram is a rectangle in the $uv$-plane

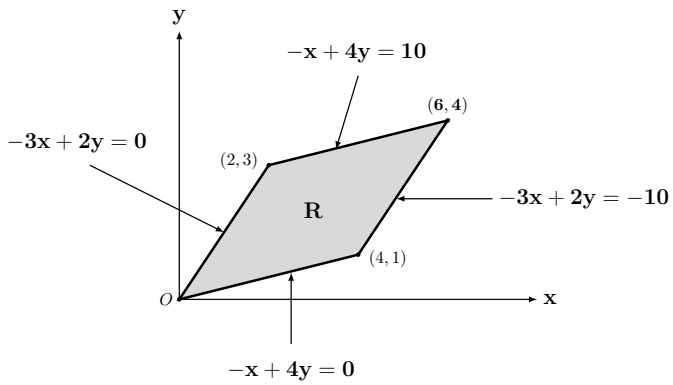


Figure 10: Equations of the Sides of the Parallelogram

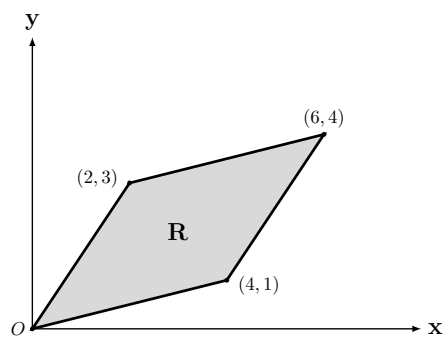


Figure 11: Parallelogram in the xy-plane

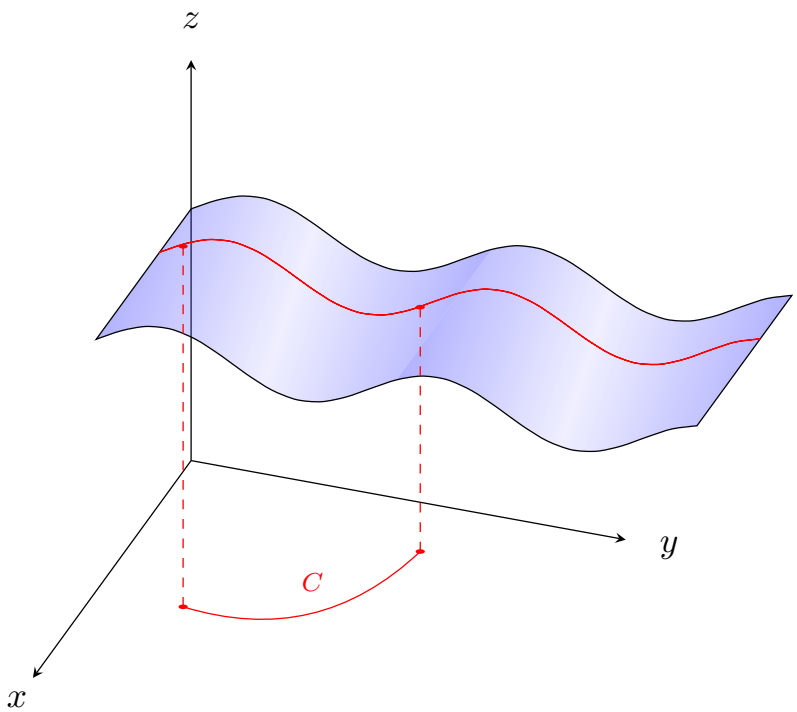


Figure 12: 3D

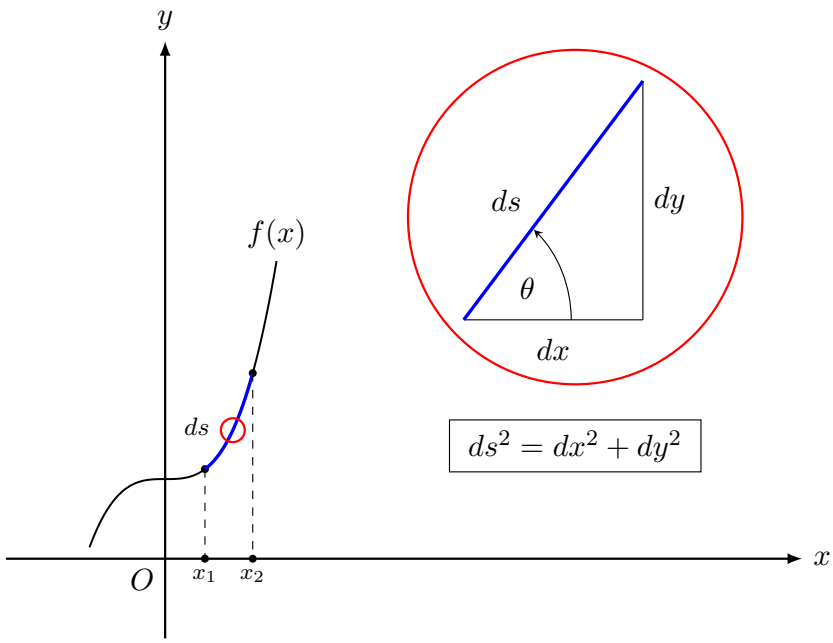


Figure 13: $f(x)$, $ds$ and the Pythagorean Theorem

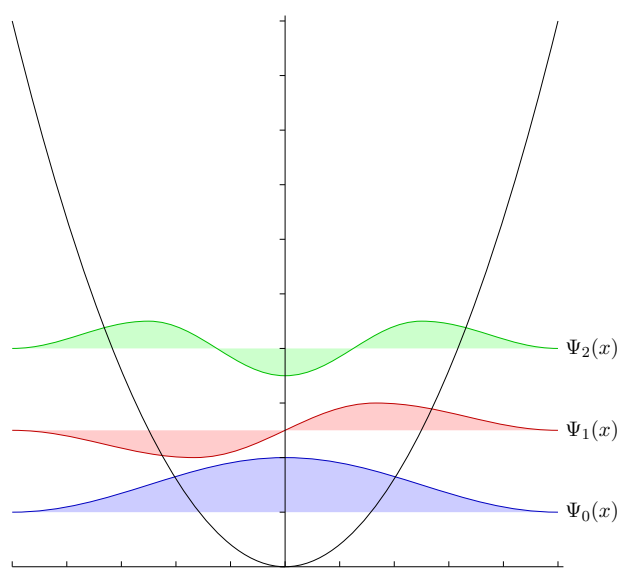


Figure 14: Wave Functions

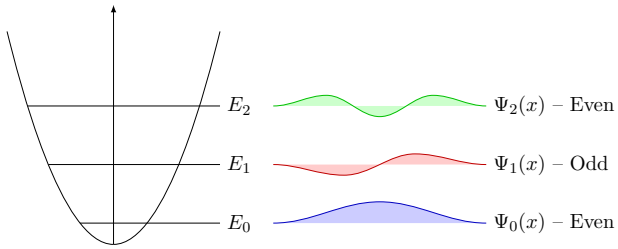


Figure 15: Wave Functions

$$
\begin{aligned}
9^2 &= 81 \longrightarrow 8+1 = 9 \\
45^2 &= 2025 \longrightarrow 20+25 = 45 \\
703^2 &= 494209 \longrightarrow 494+209 = 703 \\
7777^2 &= 60481729 \longrightarrow 6048+1729 = 7777 \\
857143^2 &= 734694122449 \longrightarrow 734694+122449 = 857143
\end{aligned}
$$

Figure 16: A Few Example Kaprekar Numbers

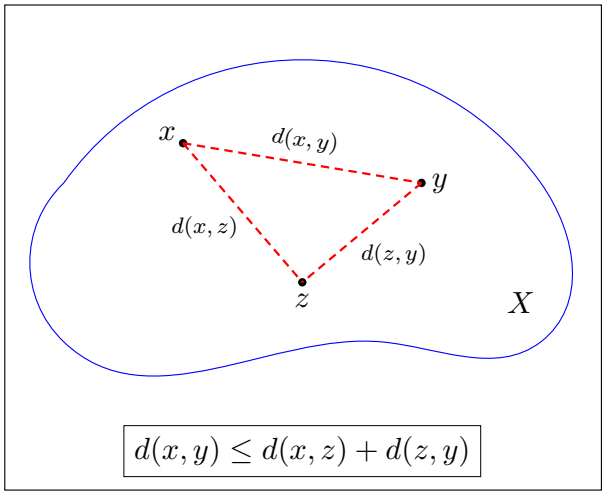


Figure 17: The Triangle Inequality

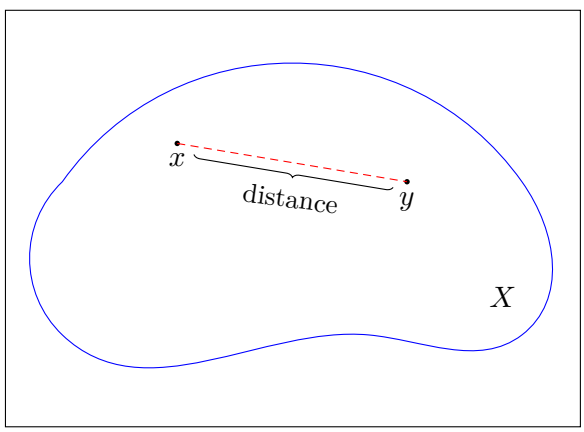


Figure 18: $X, x$ and $y$

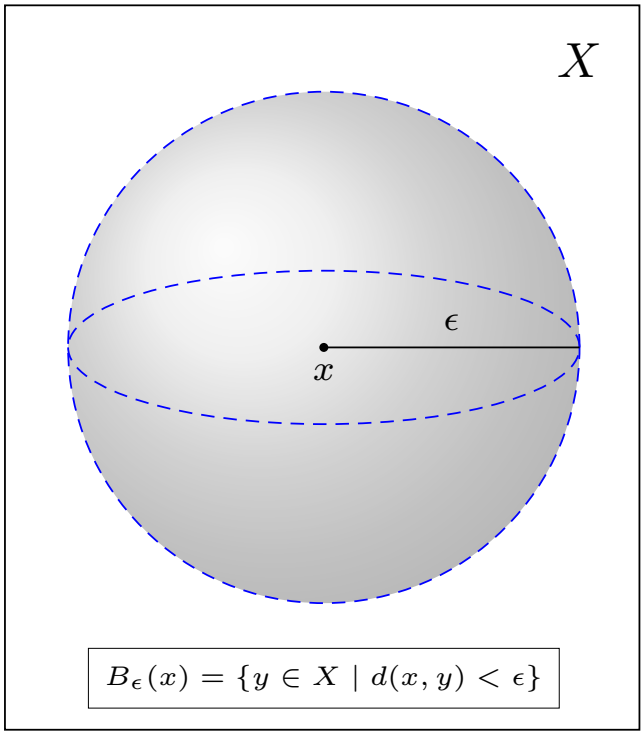


Figure 19: $B_\epsilon(x)$ is an open epsilon ball centered at $x$ with radius $\epsilon$

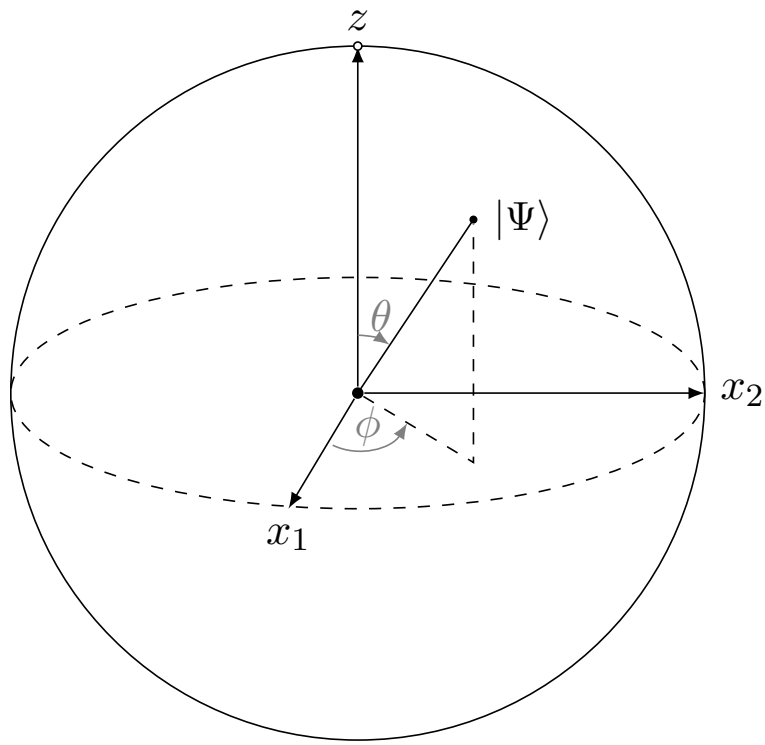


Figure 20: The Bloch Sphere

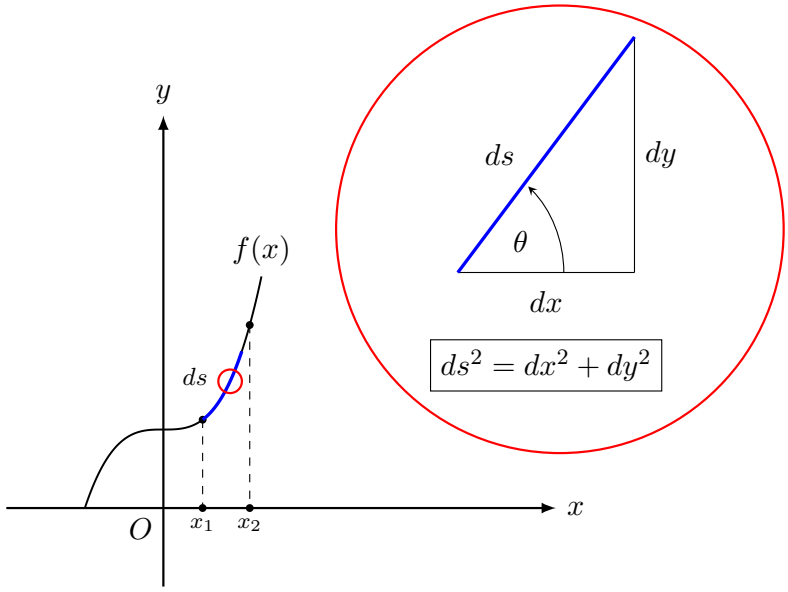


Figure 21: $f(x)$, $ds$ and the Pythagorean Theorem

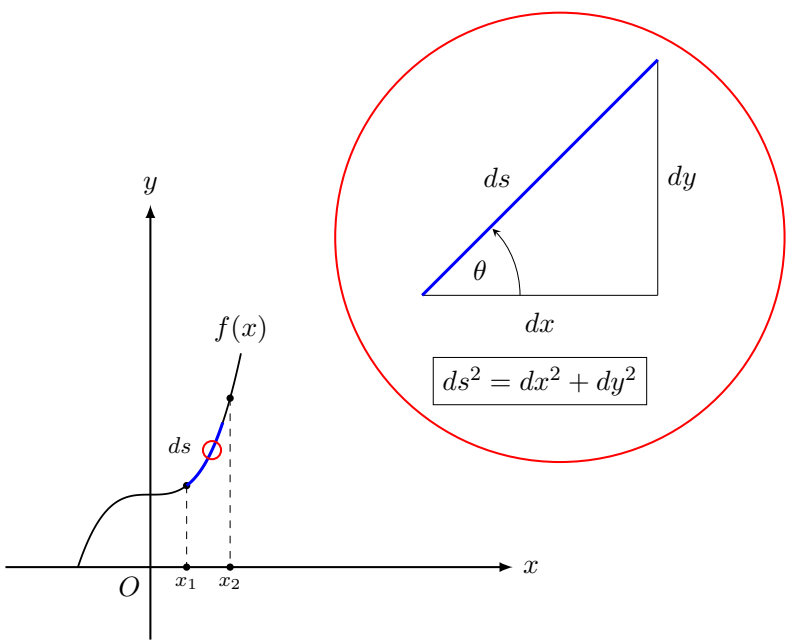


Figure 22: $f(x)$, $ds$ and the Pythagorean Theorem